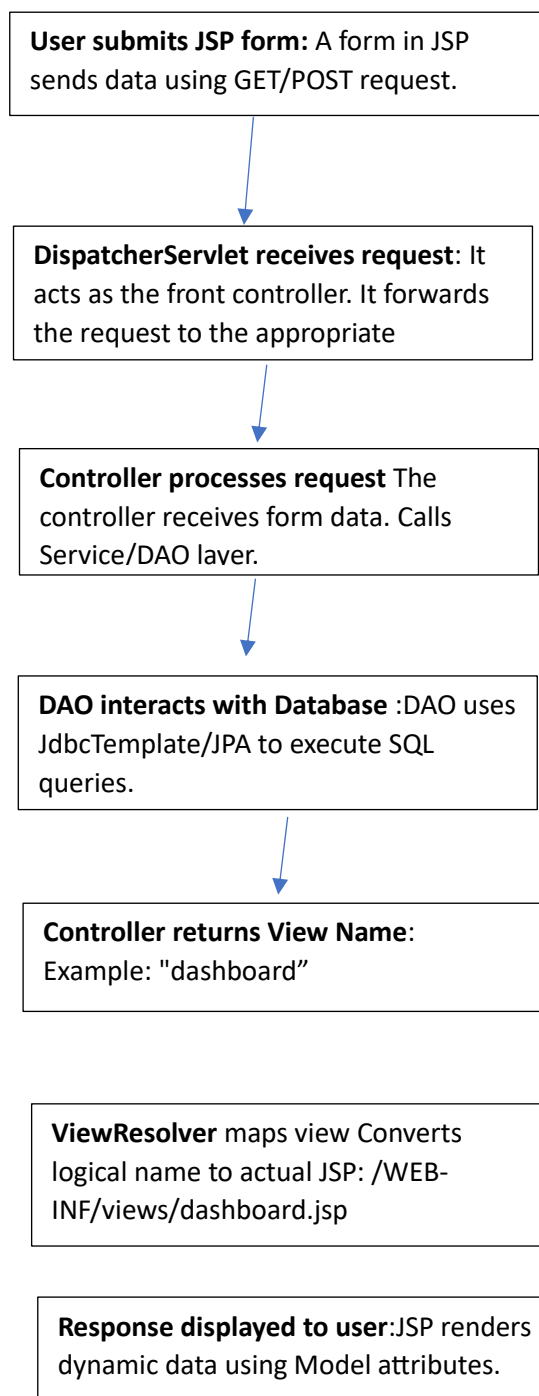


Spring

Section A: Conceptual Application

1. Explain the flow of a request in Spring MVC from JSP form to Controller to database and back to view.



2. Explain IoC (Inversion of Control) and Dependency Injection in Spring.

IoC means the Spring container controls object creation and lifecycle instead of the programmer manually creating objects.

Without Spring:

```
UserDAO dao = new UserDAO();
```

With Spring:

- Spring creates and manages objects (beans).

Dependency Injection (DI)

DI is the process of injecting required dependencies into a class.

Example:

```
@Autowired  
private UserDAO userDAO;
```

Spring automatically injects the object.

Benefits

- Loose coupling
- Easy testing
- Better maintainability
- Reusable components

3. Differentiate between BeanFactory and ApplicationContext.

Feature	BeanFactory	ApplicationContext
Type	Basic container	Advanced container
Bean Loading	Lazy loading	Eager loading
Enterprise Features	Limited	Supports AOP, Events, i18n
Performance	Lightweight	More powerful
Recommended	Rarely used	Commonly used

4. Explain how JdbcTemplate simplifies JDBC operations

JdbcTemplate removes boilerplate JDBC code.

Without JdbcTemplate:

- Load driver
- Create connection
- Prepare statement
- Handle exceptions
- Close resources

With JdbcTemplate:

```
String sql = "select * from users where username=? and password=?";  
List<User> users = jdbcTemplate.query(sql,  
    new Object[]{u, p},  
    new BeanPropertyRowMapper<>(User.class));
```

Advantages

- Automatic resource management
- Cleaner code
- Exception handling
- Reduced boilerplate

7.Explain the role of DispatcherServlet in Spring MVC.

DispatcherServlet is the front controller in Spring MVC.

Responsibilities

- Receives all requests
- Finds appropriate controller
- Handles request mapping
- Coordinates view resolution
- Returns response to client

Flow:

Client → DispatcherServlet → Controller → ViewResolver → JSP

5. Explain constructor injection vs Setter Injections

Constructor Injection	Setter Injection
Dependency injected through constructor	Injected through setter method
Best for mandatory dependencies	Best for optional dependencies
Ensures object completeness	Allows modification later
More secure	More flexible